Case: Disaggregation

Introduction[1]

Considerable research during the last two decades has concentrated on aggregate production planning, inventory control, and scheduling. *Aggregate planning* procedures, which have been proposed for both manufacturing and service organizations, help determine monthly (or quarterly) output, inventory and manpower aggregates. Inventory and scheduling procedures, using the aggregate decisions as input, tend to focus on such short-term decisions as (i) the sizing and timing of production (purchase) orders for specific items, (ii) sequencing of individual jobs (orders), and (iii) short-term allocations of resources to individual activities and operations. The total process of going from aggregate plans to more detailed plans can be called ***disaggregation***.

Disaggregation is an important issue in manufacturing as well as service organizations. Depending upon the nature of the production system, disaggregation decisions in a manufacturing organization may exist on one or more of the following three levels:

1. Given aggregate decisions on output and capacity, determine the timing and sizing of specific final product production quantities over the time horizon (sometimes referred to as a master schedule).
2. Given the timing and sizing of final product production quantities, determine the timing and sizing of manufactured (or purchased) component quantities.
3. Given the timing and sizing of component quantities, determine the short-term sequences and priorities of the jobs (orders) and the resource allocations to individual operations.

The taxonomy shown on the left side of Figure 1 provides the framework with which the three levels of disaggregation in manufacturing organizations.

Disaggregation problems in service organizations possess the complicating characteristics of stringent response time requirements, time dependent demand rates, and no finished goods

[1] Adopted from Ritzman, L.P., & Tvrdý, M. (1979). *Disaggregation Problems in manufacturing and service organizations* (1st ed. 1979. ed.). Dordrecht: Springer Netherlands: Imprint: Springer.

inventories to smooth production rates. Disaggregation decisions in service organizations also exist at three levels:

1. Given aggregate decisions on output and capacity, allocate manpower and other resources to specific operations over the time horizon (sometimes referred to as the staff sizing problem).
2. Given the allocation of resources to specific operations, determine the shift schedules and crew assignments of employees.
3. Given the shift schedule assignments, determine short-term adaptations, reallocations between operations, and priorities of the service requirements.

The right half of Figure 1 provides a basis for describing disaggregation problems in the service organizations.

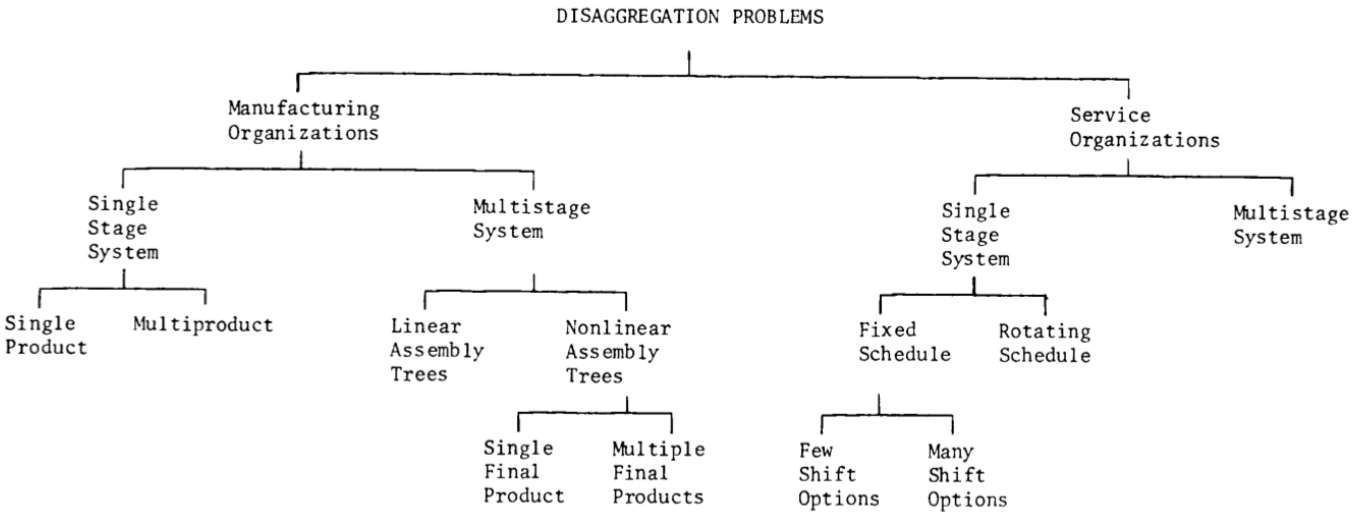


Figure 1: A taxonomy of disaggregation problem.

Watch

Watch the following videos:

- Definitions (from 41:21 onward)
- Disaggregation

Problem

Renova's problem is to develop a *disaggregate production plan* (Economic Lot Schedule) for five SKUs (white, blue, red, green, and pink) of a product to determine the optimal cycle length in order to minimize the total cost (or equivalent objective). The notations are presented in Table 1. However, you need to define new variables if required.

Table 1: Notations

Parameter/Variable	
j	Indices of set of SKUs, $j \in \{w, b, r, g, p\}$
D_j	Demand of SKU j
I_j	Inventory on hand of SKU j
P_j	Available manhours for producing SKU j
P_t	Total available manhours in cycle T
T	Cycle length

Note: The unit of measurement for $D, I, and\ P$ is manhours.

Part A (8 marks)

Demand and inventory on hand for five SKUs are presented in Table 2:

Table 2: SKU's demand and inventory on hand for Part A.

	White	Blue	Red	Green	Pink
D_j	300	900	600	700	500
I_j	200	500	500	300	100

Given that the demands are deterministic and in manhours, and the available manhours (P_t) for the period is 3000,

i. determine the sequence of producing the SKUs and show it in a simple graph.
ii. develop a linear programming model (in MS Word) and then solve it in Excel.
iii. explain the findings in details (MS Word)

iv. develop a table in which inventory position of all SKUs is presented for periods of switching the production for each SKU (MS Word and MS Excel). For inventory positions, round to the nearest integer and compute the sum of inventory on hand for each period.
v. Is maximization of T in conflict with the JIT paradigm where we try to achieve the minimum manufacturing cycle time?

Part B (8 marks)

Suppose we keep the information regarding the demand, inventory on hand, and the total production manhours from Part A. In this section, we want to avoid keeping excess inventory before the production begins. In other words, Renova starts to produce the items only when the inventory on hand is zero. You are required to

i. develop a linear programing model to provide an optimal production schedule for the five SKUs given that production starts when the inventory of an item is zero. (MS Word)
ii. Solve the problem in Excel.
iii. Discuss your findings (MS Word).
iv. develop a table in which inventory position of all SKUs is presented for periods of switching the production for each SKU (MS Word and MS Excel). For inventory positions, round to the nearest integer and compute the sum of inventory on hand for each period.
v. Explain in which model (Part A or Part B), the probability of shortages is higher. Why?. What is your recommendation?

Presentation (4 marks)

Organization, style, structure of sentences, flow of information, presenting the mathematical expressions in right format, citations and referencing, structure of tables and figures and the associated captions, and overall presentation.

Important Notes:

Note 1:

Microsoft Excel does not accept a cell to represent the objective function if there is no formula in it. In our case, we try to 'Max T' and thus we consider a cell for T. The problem is that as there is no formula in the objective function cell, you will see the following error:

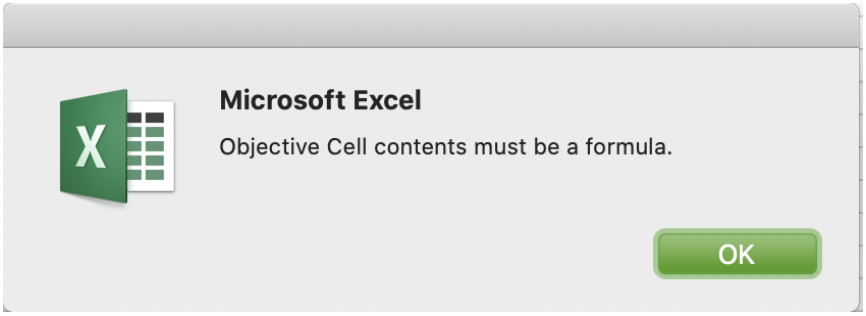


Figure 2: Error if there is no formula in the cell designated for the objective function.

To overcome this issue:

a. Consider **two** cells for the value of T.
    i. The first cell (let's call it G8) needs to be adjacent to the decision variables (if they are located in B8 to F8). When developing the left hand side of the constraints, use this cell (G8).
    ii. The second cell (Let's call it B11) is the location of objective function. Simply, add a formula in B11 to make it equal to the first cell (G8), "=G8". Do not use this cell for developing the LHS of constraints.
b. When setting up the Solver parameters, select decision variables and G8 altogether. In other words, select 'B8:G8'.
c. Select the second cell (B11) as the location of the objective function.
d. Solve the problem.

Note 2:

If you are not familiar with scientific notation of expressing large/small numbers, read this page. For example, if the optimal value of a decision variable comes up as 1.38777878078145E-17, you should consider it as zero.